GUIDE 3

TESTING USING THE MT4 STRATEGY TESTER

Copyright©2012 – Expert4x. ALL RIGHTS RESERVED. This ebook is intended for the purchaser's personal use only. No part of this eBook may be reproduced or transmitted in any form whatsoever, electronic, or mechanical, including photocopying, recording, or by any informational storage or retrieval system without express permission from Expert4x.

Contents

INTRODUCTION TO THE STRATEGY TESTER 3
SETTING THE STRATEGY TESTER UP FOR THE MAGIC MOVING AVERAGE EXPERT ADVISOR 4
THE STRATEGY TESTER TABS 9
THE MAGIC MOVING AVERAGE EXPERT ADVISOR PROPERTIES 11
TESTING THE MAGIC MOVING AVERAGE EXPERT ADVISOR 14
In summarised and analysed format (Report tab). 17
As a transaction list (Results tab) 19
As an equity movement graph (Graph tab) 20
On the chart (click on open chart), ... 21
QUESTIONS AND ANSWERS 22
Question 1: Explanation of testing record terms 22
Question 2: Using tick data 22
Question 3: Testing and optimisation speeds 23
CONCLUSION OF GUIDE 3 24

Objectives of Guide 3:

- To introduce the MT4 Strategy Tester
- To do the setup of the MT4 Strategy tester for testing and optimising
- To create an understanding on the Expert Advisor properties
- To do a basic Test of the Magic Moving Average Expert Advisor

INTRODUCTION TO THE STRATEGY TESTER

An Expert Advisor is an automated Forex trading program / robot that will trade for you without supervision. Many people see this as the Expert Advisor's main and only function. There are, however, many other benefits to having an Expert Advisor.

When running the Expert Advisor through the MetaTrader 4 Strategy Tester you can

- Test an Expert Advisor to see how it would have performed in the past.
- You can learn when the Expert Advisor does well and when it does poorly
- You can optimise your Expert Advisor to give you the best historical settings
- You can reduce the risk of trading the Expert Advisor by identifying the risk related to various settings
- You can see historical trades on a trading account format, on charts, analysed reports etc that will help you learn more about the nature of the Expert Advisor and its performance.
- And much more....

The MetaTrader 4 Strategy Tester is included with all MT4 installations. **Testing** the performance of Expert Advisors using trading history answers the question "If I had used this Expert Advisor with these settings over this period how much would I have made?" **Optimising** the Expert Advisor answers the question "Which settings would have given me the best income in the past?"

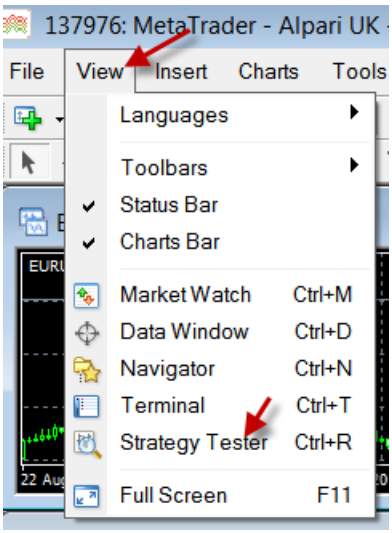
The MetaTrader 4 Strategy Tester is one of the most powerful and yet underutilized Forex Trading tools available to Forex Traders.

In these Guides we show you how to get the best from your Magic Moving Average Expert Advisor using this powerful trading tool.

By now you should have loaded the Magic Moving Average Expert Advisor into your MetaTrader4 trading installation as well as the test installation created in Guide 2. You will also have loaded the test data we are going to use. If you have not done so, do this before working your way through this guide.

SETTING THE STRATEGY TESTER UP FOR THE MAGIC MOVING AVERAGE EXPERT ADVISOR

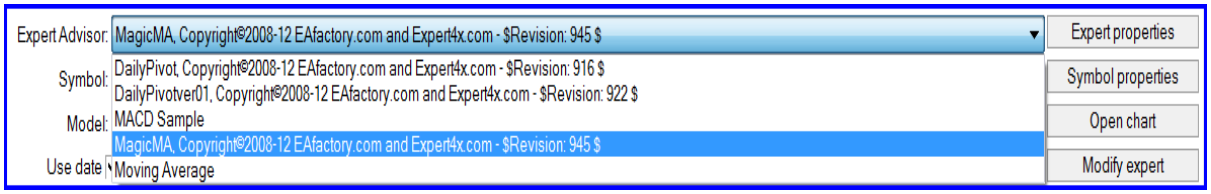
The MT4 Strategy Tester can be found by selecting "VIEW" from the main MetaTrader menu and then selecting "Strategy Tester". The short cut is Ctrl+R.



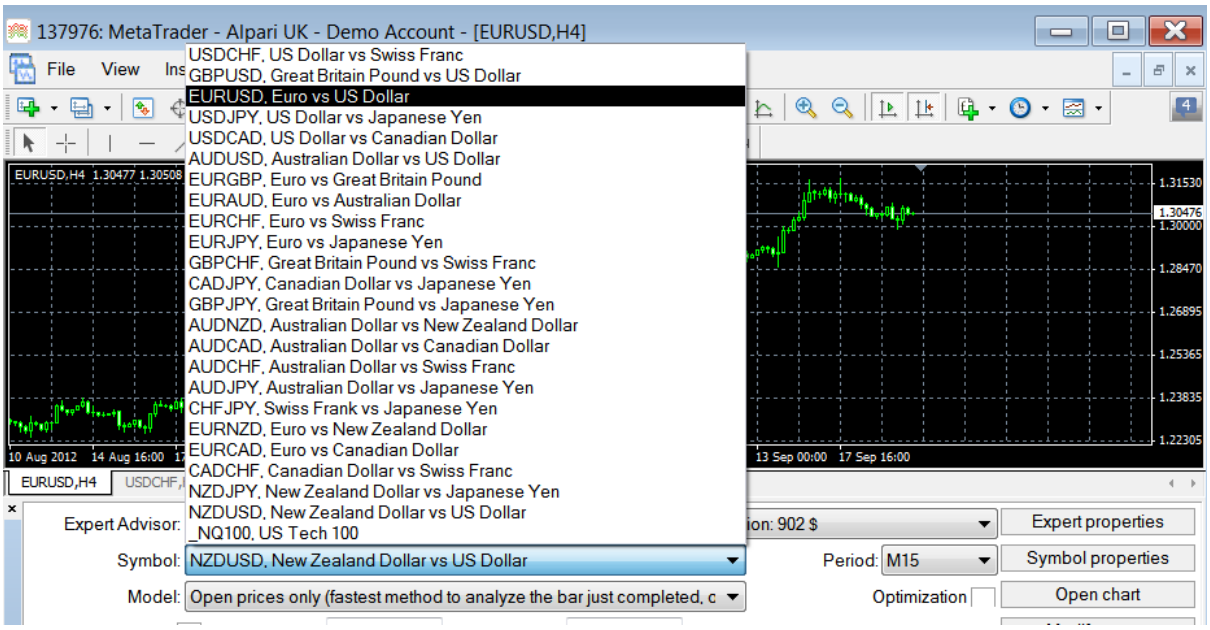
The Strategy Tester will then be displayed below your charts.

This Strategy Tester has already been setup to test the Magic Moving Average Expert Advisor so we will review each part of the Strategy Tester.

Expert Advisor: This is where you load the Expert Advisor which you would like to test and optimise. By clicking on the drop down menu it would list all the Expert Advisors on your installation and you would select the Magic Moving Average EA.



Symbol: The symbol relates to the currency cross that you would like to use for your test or optimisation. We are going to use the EURUSD. Select the currency from the drop down menu.



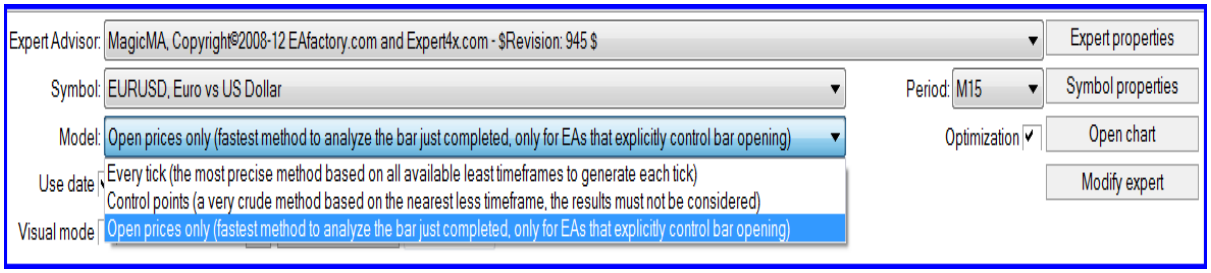
Model: Model refers to the method of testing to be used. There are 3 options given.

- Open prices only
- Control Points
- Every Tick

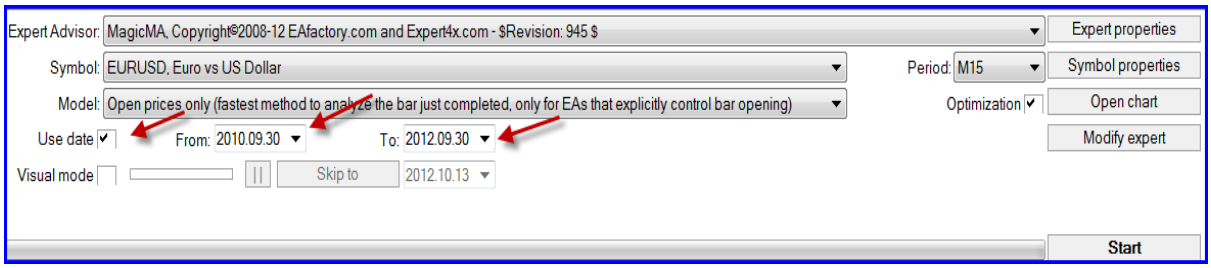
The Magic Moving Average Expert Advisor makes its trading decisions on the open of a new candle. The stops and targets are normally quite big so both are unlikely to occur in the same candle. We would therefore select the “Open price only” option.

If your Expert Advisor can enter a deal at any time and there is a likelihood of the target or the stop occurring on the same candle you would use the “Every Tick” option.

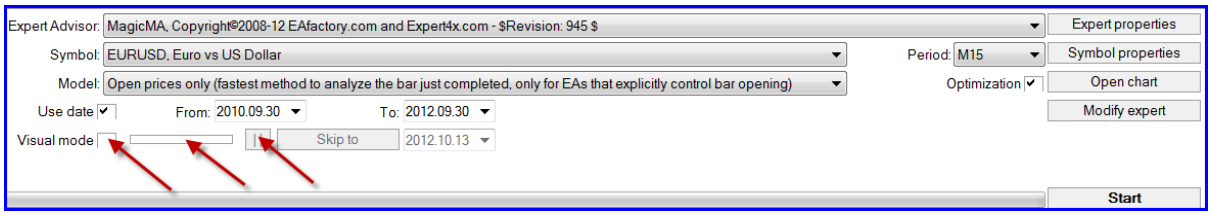
The control point option is rarely used.



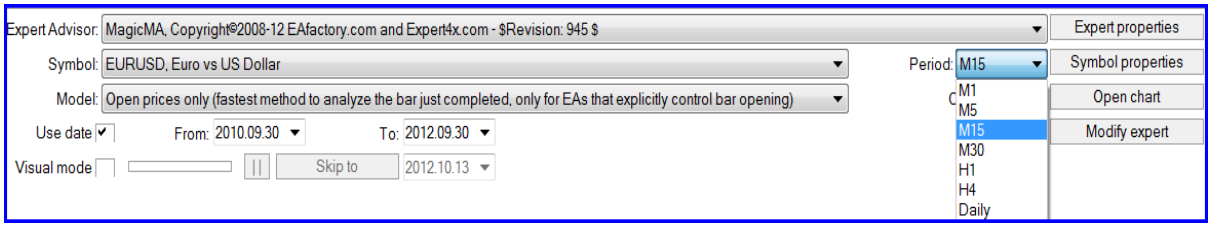
Use date: The "Use date" facility allows you to specify exactly the period you wish to test by ticking the "Use Date" checkbox and typing the dates into the "From" and "To" blocks. We are going to test the last 2 years up until the 1st of September 2012 so we would enter those dates accordingly.



Visual mode: Visual mode allows you to view the chart as the Expert Advisor is running. This is very interesting as you can see when transactions are entered into and closed. If you know the rules of your EA this is a way of making sure that it is performing as it should. The Visual mode also allows you to slow the speed of the testing down and you can even pause the testing. During Magic Moving Average process we will not be using this facility but you are encouraged to use and experience it sometime after completing the testing and optimisation process.

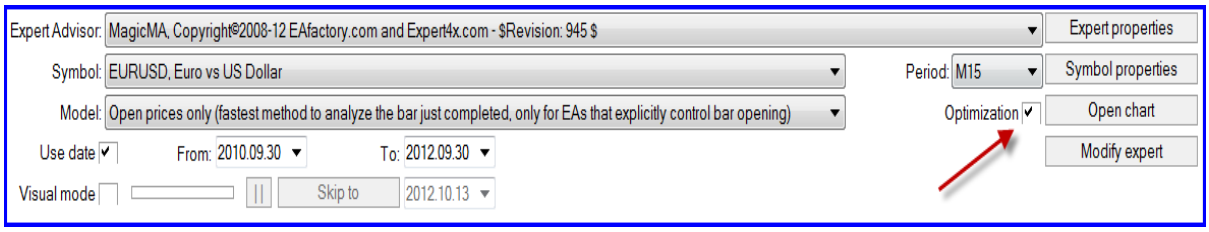


Period: Use the period to select the charts the trading will occur on. This is a very important selection as most Expert Advisors react differently in different timeframes. We are going to be using the 15 min timeframe and that can be selected from the drop down menu.

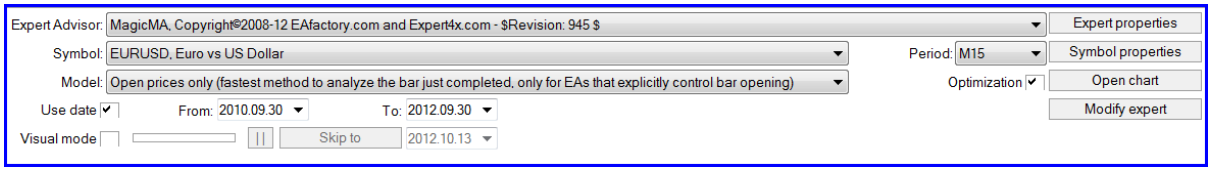


Optimisation: When this checkbox is ticked the Strategy Tester will optimise your Expert Advisor and find the settings which produce the best results. Optimisation will be covered in the next Guide.

If un-ticked the strategy Tester will test the Expert Advisor based on its current settings. We will be testing the Expert Advisor later on, so we will leave settings as they are.

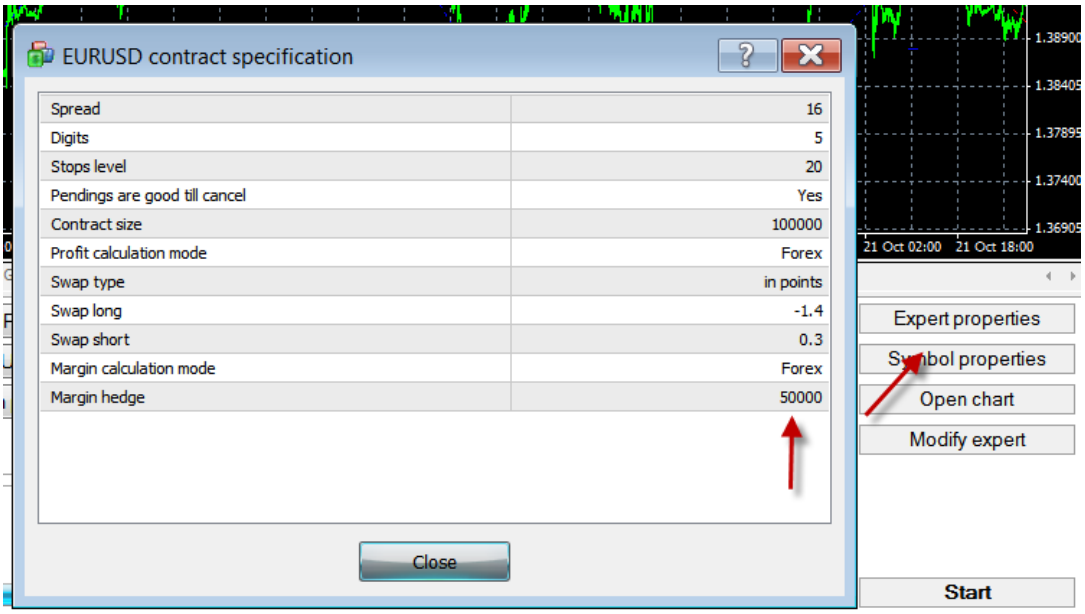


Modify Expert: This button is used by developers of Expert Advisors to amend the EA directly from the Strategy Tester as a result of the testing performance. We don't have to worry about this option.



Open Chart: After you have performed a testing activity you can view the trades generated by the EA on the trading charts by click on this option.

Symbol properties: This shows you the symbol properties used for the test. You can, for instance, use this to change the spreads or gearing of the currency you are testing. All of our testing is done offline so use this if you want to change any of your currency properties. Initially it is best to use the default setting and change these when doing final refinements.



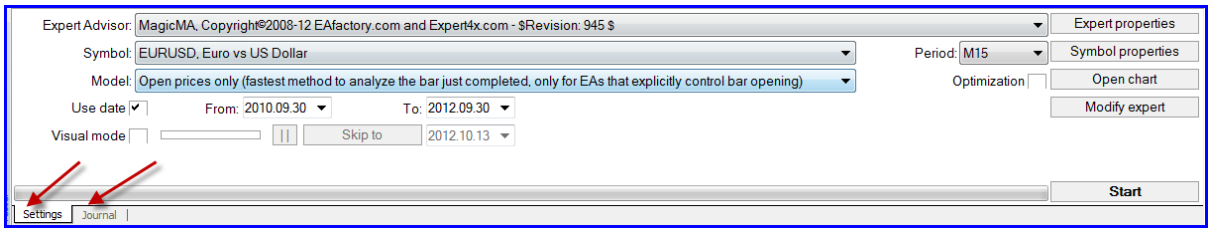
THE STRATEGY TESTER TABS

Before testing: there will only be two tabs visible.

- Settings tab
- Journal tab.

The Setting tab shows the main setup screen that we have been working on.

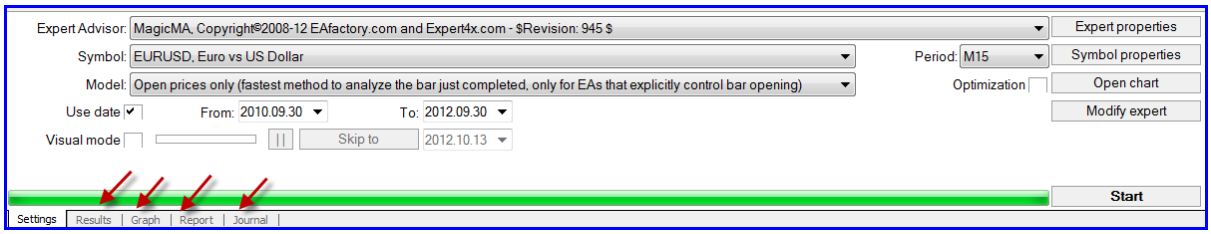
The Journal tab creates a log of activities that have occurred while using the Strategy Tester. At this stage we don't need to refer to this tab.



After testing: There will be 3 tabs added

- Results
- Graph
- Report

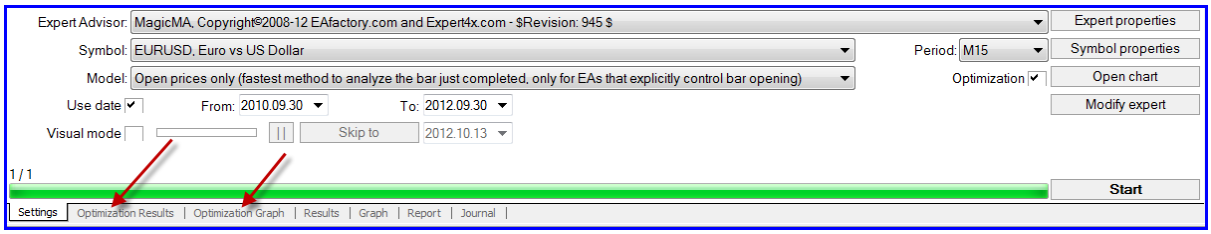
We will go into the detail of these tabs once we have run our first test.



After optimisation: There will be 2 more tabs added

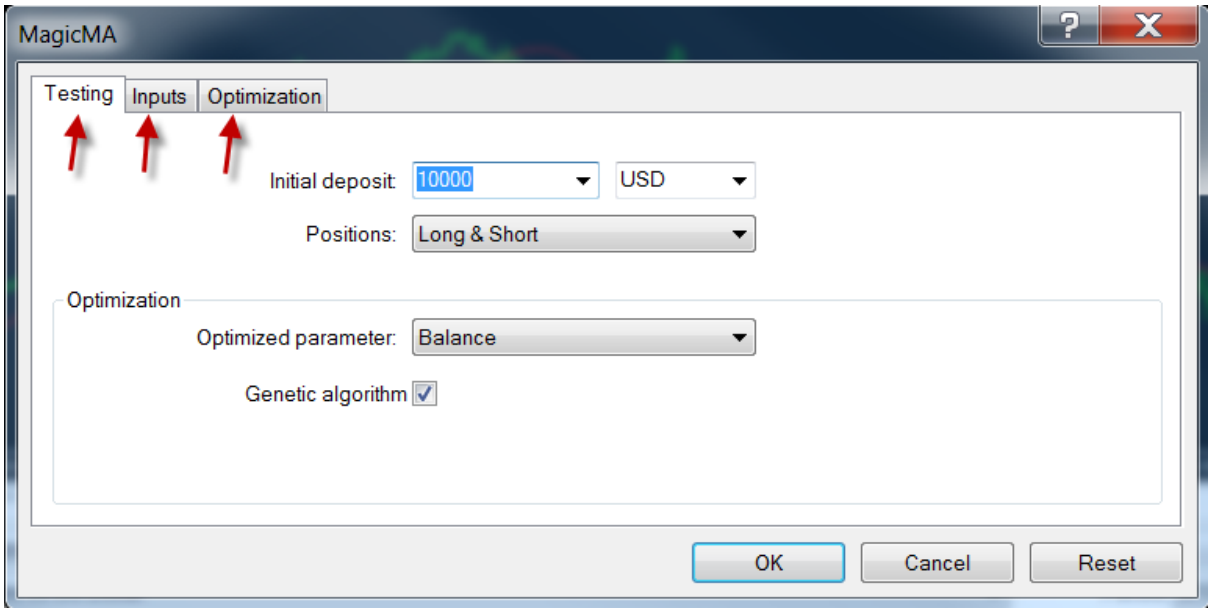
- Optimization Results
- Optimisation Graph

These will be covered in more detail in Guide 4 on optimisation

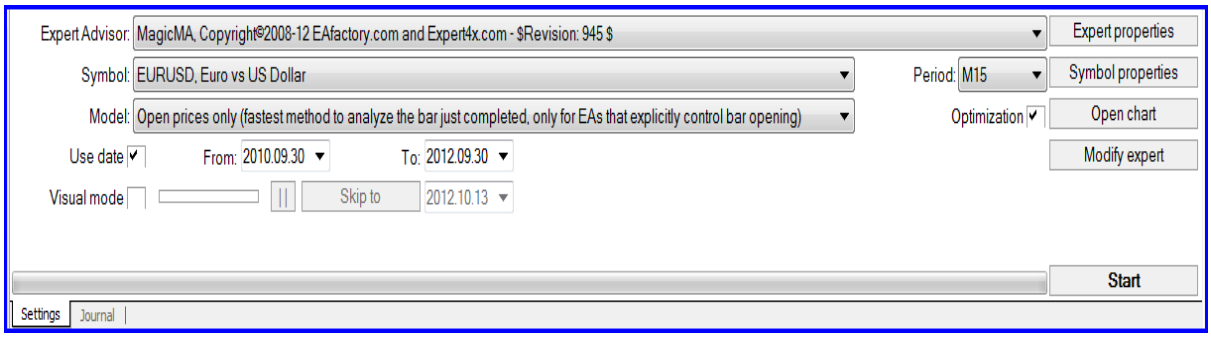


Start button: This is used to start a test. This will also start the optimisation process when the Optimisation block is ticked. Depending on a number of factors (refer to Guide 4) tests and optimisations can run for a few seconds to a number of hours. Tests run very quickly as only one alternative is processed. Optimisation can process millions of alternatives in one run and that will take more time.

Expert Properties: The Expert Properties button opens the actual details of Expert Advisor that you have loaded. We will cover this in the next section.



This concludes the setting up of the MT4 Strategy Tester. Your setup should look like the screen below at this stage. We will now move on to the settings specific to the Magic Moving Average Expert Advisor.



THE MAGIC MOVING AVERAGE EXPERT ADVISOR PROPERTIES

As shown in the previous section by clicking on the "Expert properties" button a specific DailyPivot properties window is opened. There are 3 tabs to this window and we will work our way through these 3 tabs

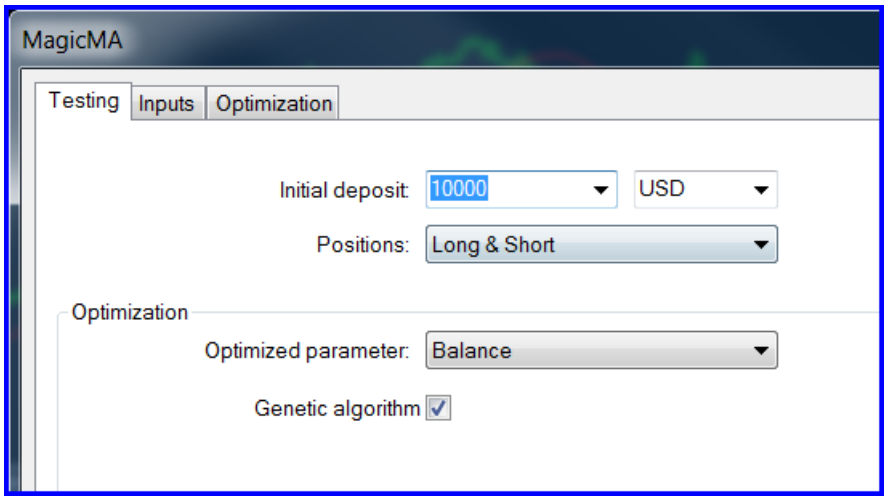
The testing tab: This tab sets the parameters used for testing and optimisation.

Initial deposit: $10 000 in USD is a good basic testing and optimisation ***account balance*** to start with. As a refinement once you have optimised your EA settings, you may want to run a final test using the account balance that you are most likely to use for live trading.

Positions: This allows you to set the testing and optimisation process requirements for: 1) trade Sells only; 2) Trade Buys only or 3) Trade both types of transactions. Normally the EA will allow this choice but if it doesn't this is a handy facility. If the market is trending this is a quick way to ensure 'with the trend' trades only. For the Magic Moving Average EA, select Long & Short.

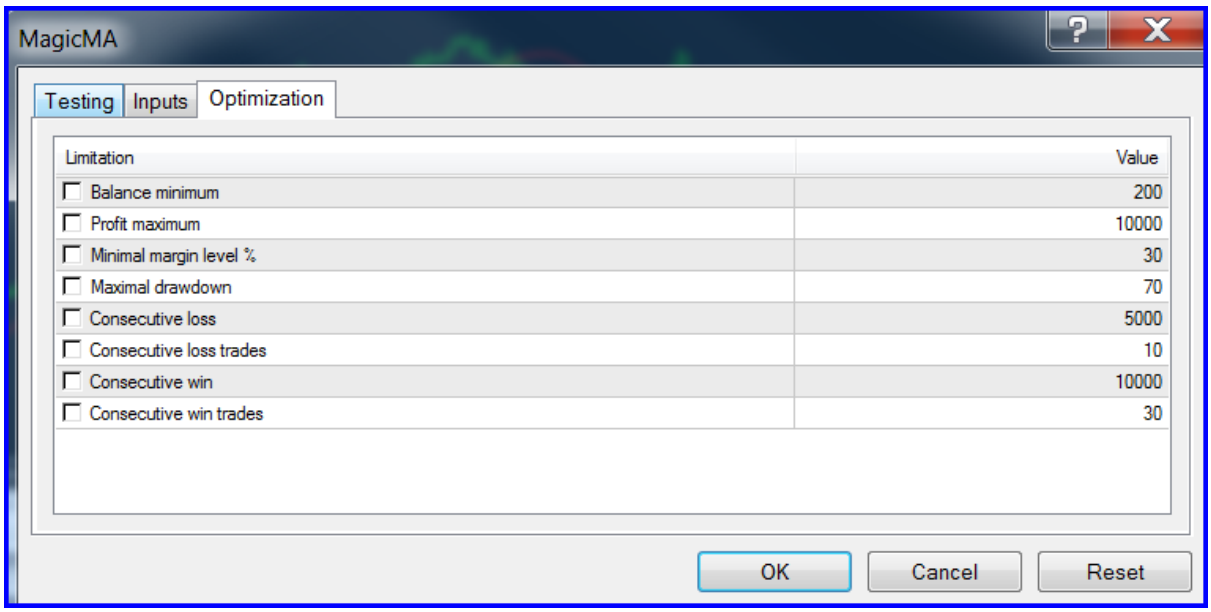
Optimized parameter: We will be optimising for the best improvement of our Account balance so select that option. Details of the other options will be discussed later.

Genetic Algorithm: This is important for optimisation as this algorithm ignores options that are clearly unprofitable. It allows for optimisation to occur much faster without a loss of quality. We would like this ticked.



The Optimisation tab: This is used if you want to place limitations on your optimisation. Once the limitation is reached the optimisation process will stop. As we do not want to place limits on our optimisation runs we can ignore this tab. Only ticked items will be activated.

Under advanced optimisation one would for instance tick Maximal Drawdown and enter 20%. The optimiser will then ignore options that produce drawdown's in excess of 20%. This speeds up the optimisation process.

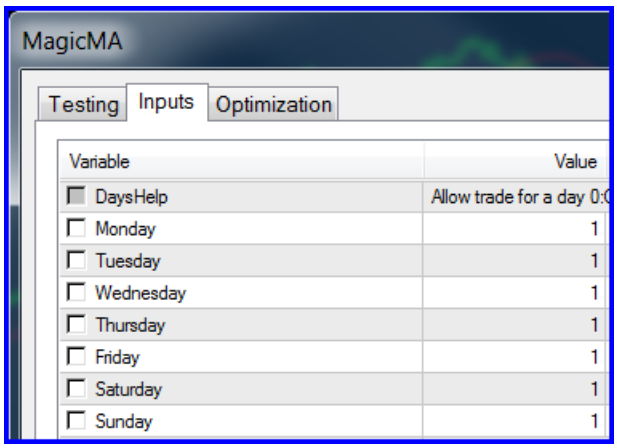


The Inputs tab: The input tab contains all the variables used by the Expert Advisor that the EA owner is allowed to vary. Information on this tab can be used for testing purposes and optimisation purposes. There are 5 columns in this tab. They will be reviewed in the discussion that follows.

The Variable column: This column contains the entire list of variables that the user can change when running this EA. The variables are covered in detail in the user guide so we will not cover them here. You will notice that there is a checkbox next to each variable. When this checkbox is ticked it will activate the variable for optimisation. If un-ticked it will be ignored for optimisation. All variables will be taken into account when doing test runs.

The Value column: The user will enter the value the Expert Advisor must use when running tests in this column. The default settings for the Expert Advisor are normally already in this column.

Only the variable and value columns are used for testing. As an example the settings below are telling the Expert Advisor to open trades only during the hours of 0 to 24 hours on Mondays to Fridays. For an explanation of the other variables please consult the user guide.



Later on we will be running the Expert Advisor using the default settings and all of this will become clearer.

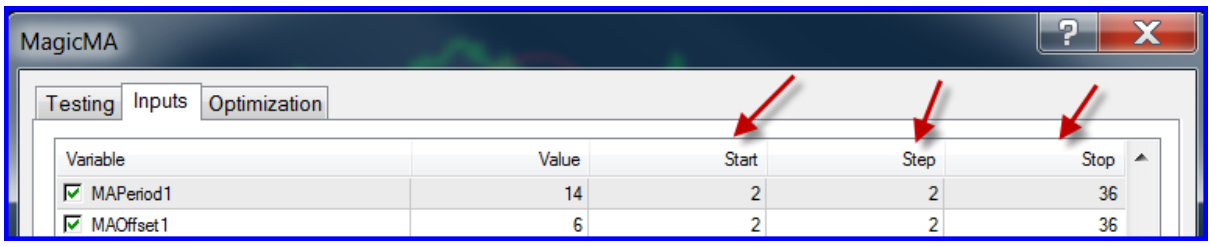
Optimisation columns: Start, Step and Stop.

The next 3 tabs are very important from an optimisation point of view. When we optimise the EA we tell the Expert Advisor which variable we want to test by ticking the box next to the variable name.

We then need to establish **the range** of the variable we need to take into account during the optimisation. We are going to use the variable ("MAPeriod1") as an example. When trading the MA period it should in generally be more than say 2 periods. We would also not like MAPeriod1 to be bigger than say 36 periods in this example. That is a rough way of determining the range that possible optimised stop sizes can be. For this range we could use 2 as the start setting and 36 as the stop setting.

The next item to be decided on is the size of jumps or steps between 2 and 36 that we want the EA to use to optimise the settings. We could decide on steps of say 2 periods. This would mean that the expert advisor would test 2 then step 2 to 4 and then step 2 to 6 and so on. The EA would step up until it reaches 36.

That is how the columns Start, Step and Stop are used in the optimisation phase in Guide 4. Please refer to this guide for more details on the optimisation process.



TESTING THE MAGIC MOVING AVERAGE EXPERT ADVISOR

We are now ready to test the Magic Moving Average expert advisor. We will be using the setting below at first. Please enter them in the value column on the inputs tab in the MagicMA properties. They are:

These setting trade the Magic MA continuously (0 hours to 24 hours) on all trading days. The second window is disabled and no settings have been entered for 2nd period. We are specifying a take target of 500 pips, although the moving average crossover will provide the close for most if not all profitable trades.

Variable	Value
DaysHelp	Allow trade for a day 0
Monday	1
Tuesday	1
Wednesday	1
Thursday	1
Friday	1
Saturday	0
Sunday	0

Variable	Value
LHelp	Lot size of market orde
Lots	0.1
TW1Help	First trading window in
ServerHourFrom1	0
ServerMinFrom1	0
ServerHourTo1	24
ServerMinTo1	0

Variable	Value
MAP1Help	Moving Average Period
MAPeriod1	65
MAOffset1	40
TP1Help	Set target profit in pips
TakeProfit1	500
TS1Help	Set trailing stop in pips
TrailingStop1	0

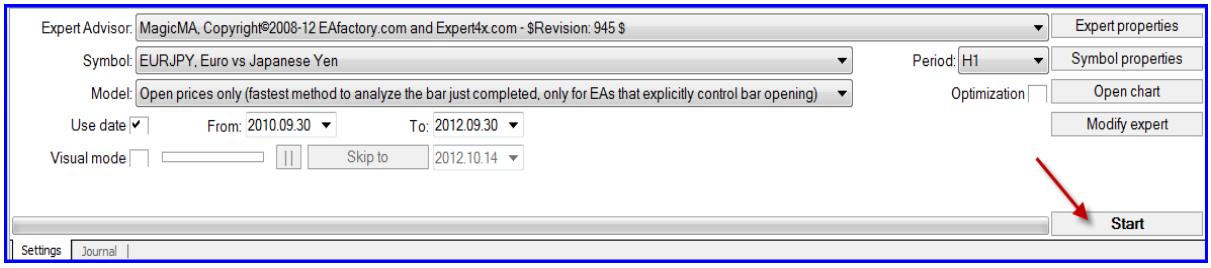
Variable	Value
SWEHelpHelp	Activate 2nd trading wi
SecondWindowEnabled	0
TW2Help	Second trading window
ServerHourFrom2	0
ServerMinFrom2	0
ServerHourTo2	0
ServerMinTo2	0

Variable	Value
MAP2Help	Moving Average Period
MAPeriod2	0
MAOffset2	0
TP2Help	Set target profit in pips
TakeProfit2	0
TS2Help	Set trailing stop in pips
TrailingStop2	0
NBWSHelp	Max delay to trade at n
NewBarWindowSec	60
ESAHelp	Broker account type E
ECNorSTPAccount	true
MagicNumber	20120930

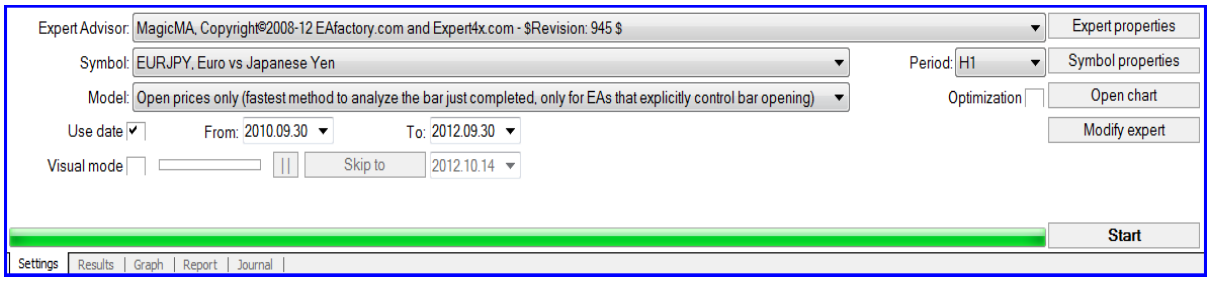
Please note that the variables are un-ticked for testing purposes. They are only ticked when during the optimisation phase which we will cover in Guide 4.

We will run this test on the 1 hour time frame. We will run the test over 2 years from 30/9/2010 to 30/9/2012. This will cover all the trading phases the currency has moved through – up and down trends and sideways markets. We will use the EURJPY which was one of the most volatile currencies over the last 2 years.

To run the test, simply click on start (making sure the optimisation block is un-ticked).



The Test will run for a few seconds or a few minutes, depending on the speed of your computer. The progress bar will turn blue on completion and 3 new tabs will appear. The Results, Graph and Report tabs.

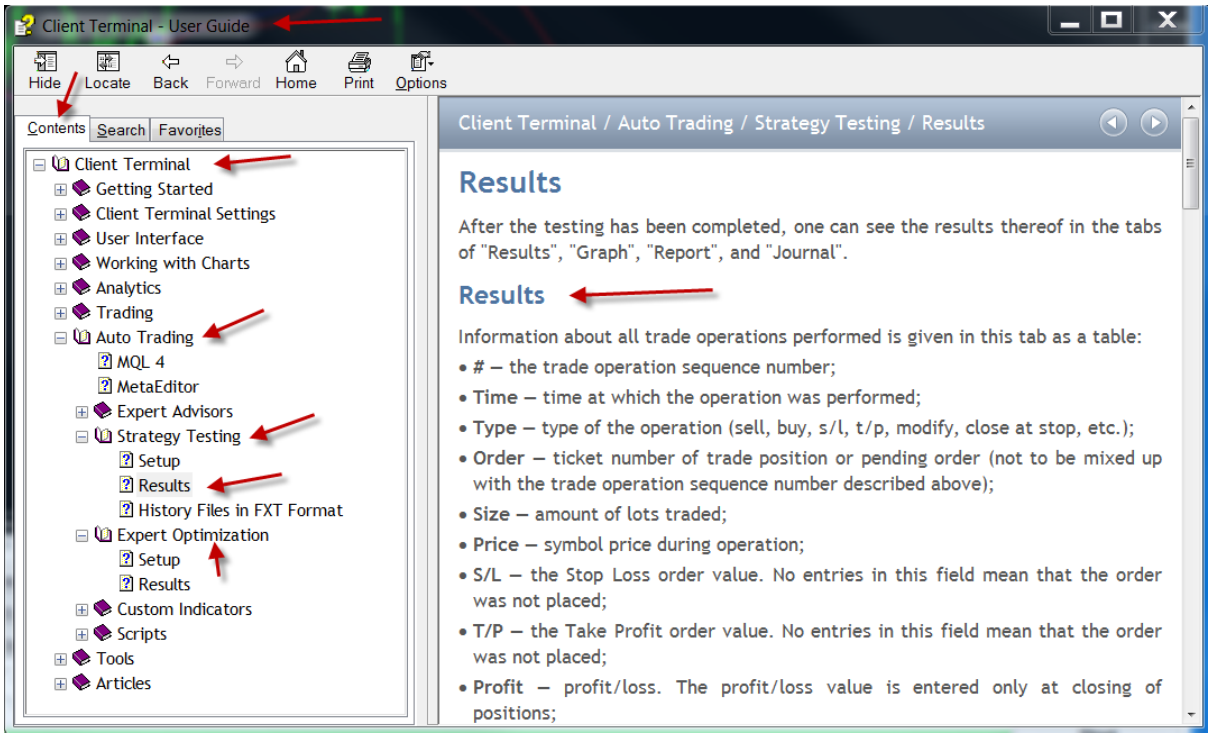


So we can view the results in 4 formats –

- In summarised and analysed format (Report tab).
- As a transaction list (Results tab),
- As an equity movement graph (Graph tabs)
- On the chart (click on Open chart),

We will discuss these formats one by one:

For more information on the meaning of certain items and descriptions please click on “Help” (or use the F1 button) and then select “Results” under “Strategy Testing” under “Auto Trading” in the “Client Terminal” section of the help contents



You will see there is help on most topics under Auto Trading. There is also information on “Expert Optimisation” which will be covered in Guide 4.

In summarised and analysed format (Report tab).

This is the quickest way of getting overall results of the test. The picture below shows the results and they will be discussed in order of importance.

Bars in test	12851	Ticks modelled	24702	Modelling quality	n/a
Mismatched charts errors	0				
Initial deposit	10000.00				
Total net profit	3842.26	Gross profit	13050.91	Gross loss	-9208.65
Profit factor	1.42	Expected payoff	10.30		
Absolute drawdown	1046.05	Maximal drawdown	1270.18 (12.42%)	Relative drawdown	12.42% (1270.18)
Total trades	373	Short positions (won %)	187 (19.25%)	Long positions (won %)	186 (19.89%)
		Profit trades (% of total)	73 (19.57%)	Loss trades (% of total)	300 (80.43%)
	Largest	profit trade	637.29	loss trade	-182.05
	Average	profit trade	178.78	loss trade	-30.70

Settings | Results | Graph | Report | Journal

Total Net Profit: This is the first information a tester wants to see – has the EA made a profit? In this case it has made one of $ 3812. Whether this is good would depend on your expectations and other factors which we will mention later on. Our expectations are currently 300 pips a month. So for 24 months this should be +/- 159 pips. This is pips is way below our expectations but guide 4 will help optimise the EA to a better production.

Mismatched chart errors: there are no errors – this confirms that we are using good test data. If you have errors please repeat the process detailed in Guide 2.

Profit factor: 1.42 - this is the ratio of Gross profit ($13 050) divided by Gross loss ($-9 208). In general as a guide this ratio should be at least 1.5 and preferably over 2. So these results are still not at the level we would like then to be.

Absolute / maximal drawdown: Absolute drawdown is the amount your account went down from the start balance. Maximal drawdown in the maximum amount your account went down from a high balance account peak. Based on the profit figure this is high but based on the account balance draw downs of under 10% are acceptable.

Total trades: 373 trades over a 2 year period using the 1 hour timeframe with big MA periods seems fine.

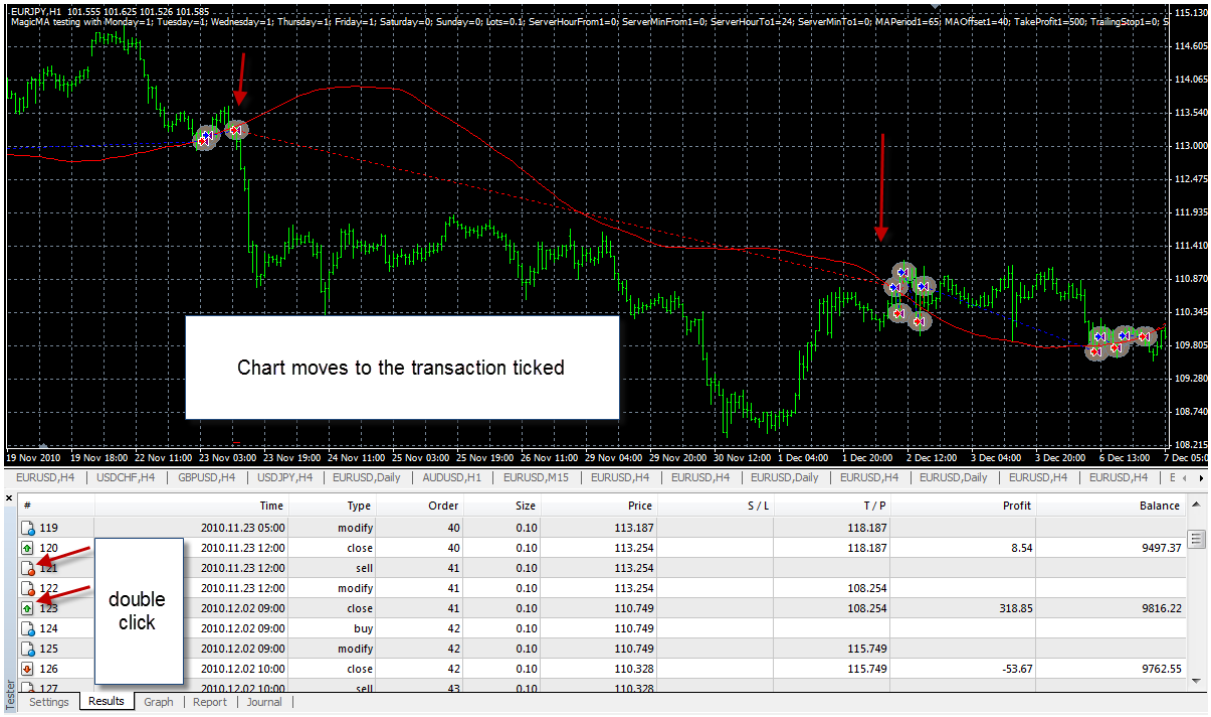
Profit trades: This gives the success rate. 19% says that only one fifth of trades are successful.

Average profit and loss trade: the average profit made on a winning trade is $178 and the loss on a losing trade is $-30. This is very good and explains why a profit was made in spite of a low success rate.

The other information helps understand the performance of the EA a little better. The general message is that the results are such that we can't go live with the default settings and that we would have to optimise this EA to achieve better results.

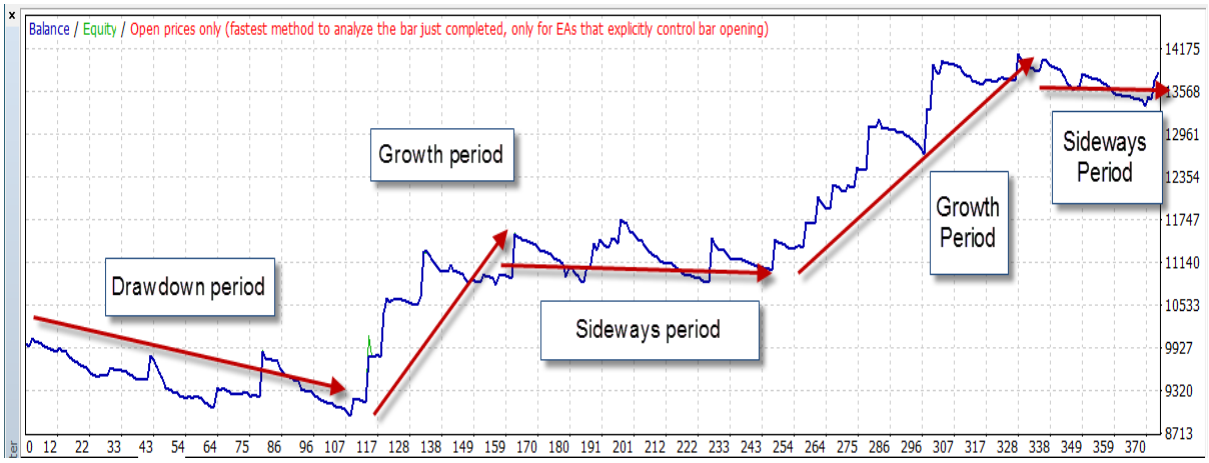
As a transaction list (Results tab)

This view shows the actual transactions one by one in date order as they would appear on your trading account. If you have the chart open and double-click on the transaction number the system will automatically take you to that specific transaction on the chart.



As an equity movement graph (Graph tab)

This graph shows the movement on your trading account. We have added arrows to show the drawdown periods and growth periods. When doing advanced optimisation it is good to try to understand why draw downs and growth periods happen.



On the chart (click on open chart),

When using this option you can view the transactions as they occurred on the charts.

QUESTIONS AND ANSWERS

Question 1: Explanation of testing record terms

Can you explain the 3 different Drawdowns, namely Absolute, Maximal and Relative? Which one will be the highest drawdown at any one time?

Answer 1: Explanation of testing record terms

Yes – for an explanation click on "Help" in the Metatrader4 main menu. Then enter "drawdown" in the "search field" and select "Strategy Tester – Results" from the search results. Your will find your answer on that page.

Run a test on the EA on the strategy tester to find out which one has the highest drawdown.

You can find the same information by clicking on "Help" then selecting "Contents" then "Auto trading" then "Strategy Testing" and then "Results".

Question 2: Using tick data

I cannot run tests on tick data. It seems the strategy tests will run reliably only on open candle. Given this, I wonder if this will run at all during normal hours? Are there forward tests or ongoing profiles I can refer to in MT4i or somewhere else? At any rate, I am going to load this EA in some demo accounts. I ran tests using ……. None of them worked with tick data in strategy tester.

Answer 2: Using tick data

Using tick data for this EA is inappropriate and therefore not mentioned in this course. I am not surprised the test does not work. The Trade decision is not made on a tick data basis. It is made on a close of the previous candle basis. So, on the close of a candle the EA looks if the price has crossed over the MA and then decides whether it will trade. If not the next time the EA is required is at the end of the next candle.

The other time tick data maybe required is for target levels that can occur in the same candle. This is unlikely as you will see that in general targets are quite large.

Beware of trying to perfect the imperfect and being distracted. It looks like you are not seeing the wood for the trees. It would be better using your time to find a more affective testing approach like making sure the testing period is appropriate to the current market conditions or using more refined heat map analysis. You will add 100's of % more value using your trading common sense than trying to perfect settings that are based on an inappropriate testing period. Remember any settings you get using tick data or close of candle data are only guides for the future and are flawed - we as Forex traders still have to add value to those guides.

Question 3: Testing and optimisation speeds

In your webinars and videos your optimisation runs and testing runs go so fast. You must have a very powerful computer?

Answer 3: Testing and optimisation speed

There are a number of factors impacting your testing and optimisation speeds. Much of the processing done for the course was done on a 3 year old laptop. Computer speed does play a role but not a major one.

Firstly: you need to be testing using the "Open Prices Only" option when using the Strategy testing – see page 6. To use this option make sure that your EA makes major decisions on the close of the candle. This will increase speed considerably

Secondly: You need to be using the Genetic Algorithm by having this option ticked in the "Testing" tab of your EA properties. This will make a major impact on processing speeds.

Thirdly: Your computer memory (RAM) should be a clean as possible. To do this merely delete all your temporary internet files via your browser or by using the "Disk Cleanup" function. Then reboot your computer before testing and optimising. This will speed your computers processing speed up.

CONCLUSION OF GUIDE 3

This completes this guide and hopefully we have achieved our objectives. Please bear in mind that we cannot explain everything about testing and optimising EA's. **We have concentrated on the areas required to get the job done**. The more you use the processes described the more they will make sense and the more you will learn. Please do not expect to be an expert and know everything there is to know about this topic in a day.

Objectives of Guide 3

- To introduce the MT4 Strategy Tester
- To do the setup of the MT4 Strategy tester for testing and optimising
- To create an understanding on the Expert Advisor properties
- To do a basic Test of the Magic Moving Average Expert Advisor

Please move on to GUIDE 4 for the optimisation phase.